

A precise checkpoint, then a useful next step

Learning intention

Apply selected ideas about hormonal control, circulation and exchange, then use evidence to choose a precise next step.

Five questions, 20 core marks.

Short recap first; 18 minutes for independent work.

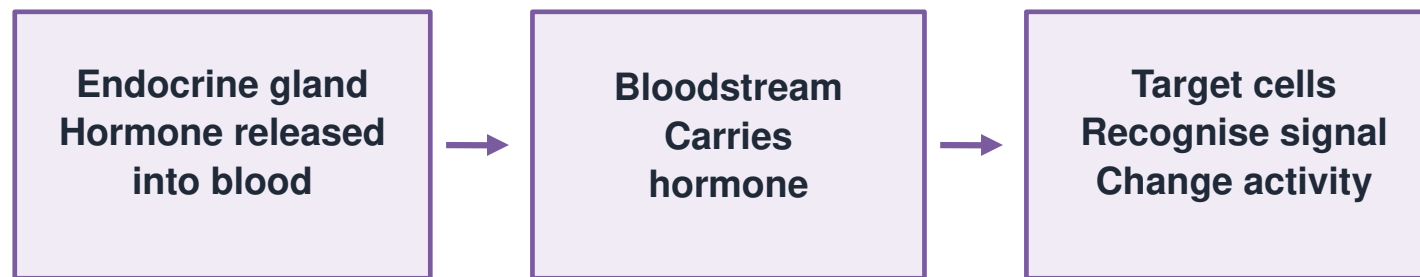
Retrieve a route and a process

R1. What carries an endocrine hormone around the body?

R2. What is the difference between ventilation and diffusion?

Trace the hormone message

One message, three important locations

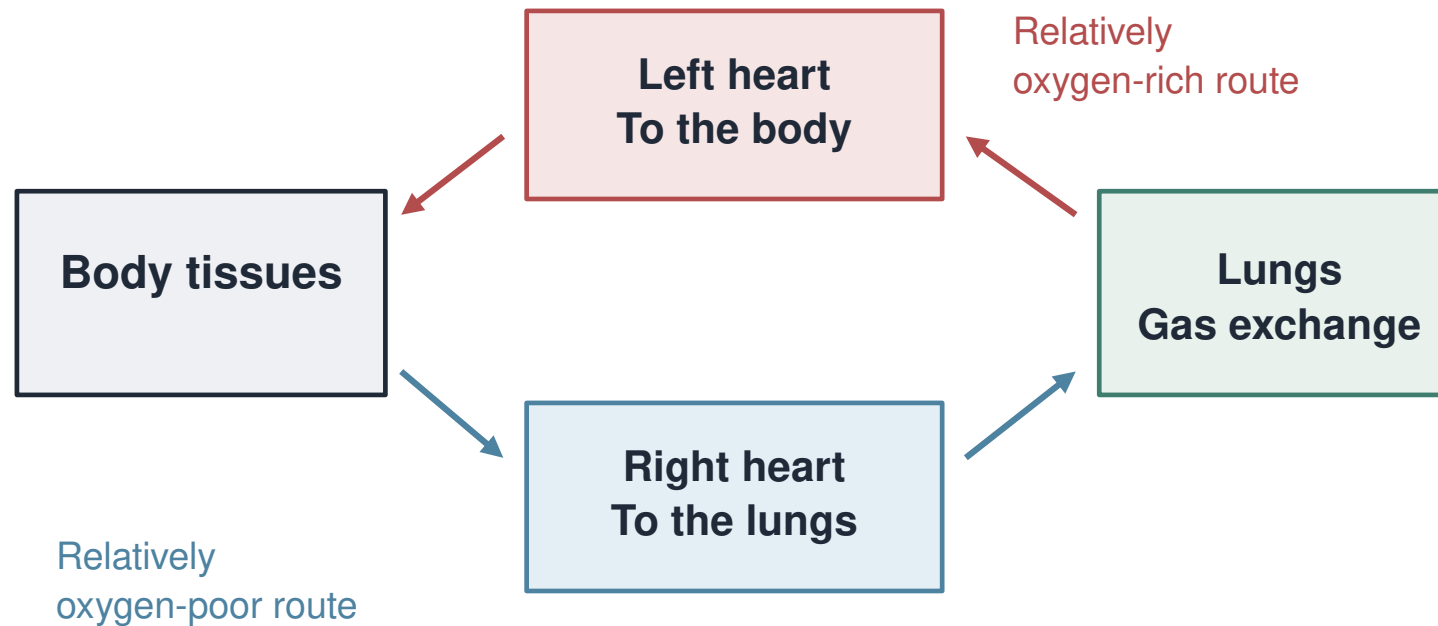


The bloodstream distributes the signal.
Cell recognition helps determine the response.

Gland releases a hormone into blood. Responsive cells recognise the signal and change activity.

Read direction before oxygen content

Two linked circuits, one continuous route

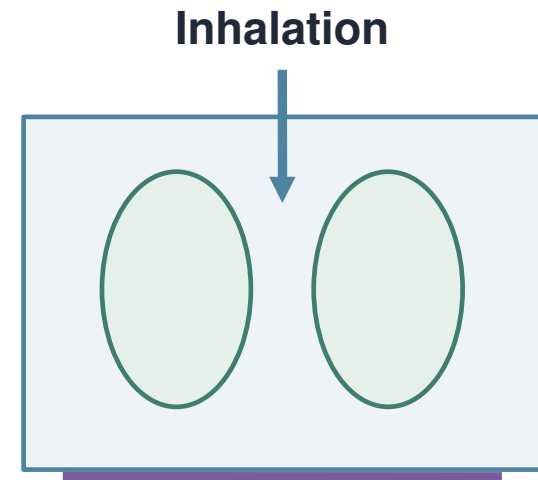


Artery = away from heart; vein = towards heart.

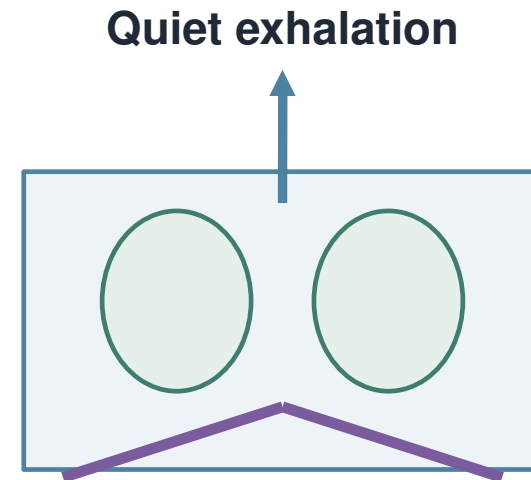
An artery takes blood away from the heart. Follow the lung circuit to test the definition.

Volume and pressure explain ventilation

Ventilation changes volume and pressure



Diaphragm contracts/flattens
Volume rises; pressure falls
Air enters



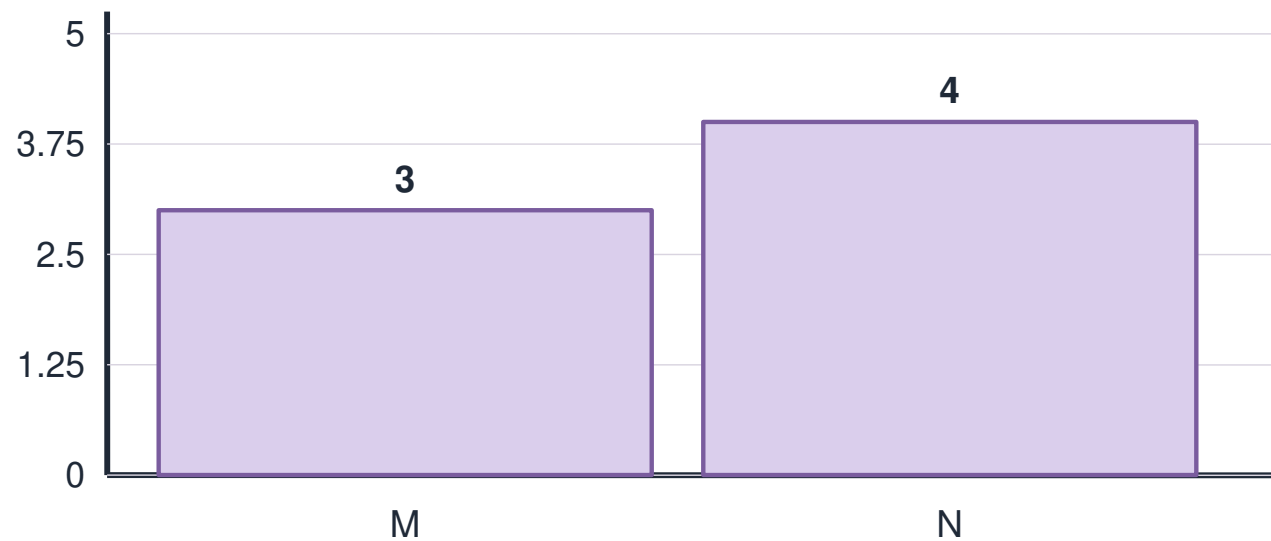
Diaphragm relaxes; recoil
Volume falls; pressure rises
Air leaves

Inhalation increases volume and lowers pressure. Quiet exhalation involves relaxation and recoil.

Use an analogous calculation

Warm-up only: model pump output

Constructed model pump output (L/min)



W1. Model M: 60 beats/min $\times$ 50 mL/beat. Model N: 80 beats/min $\times$ 50 mL/beat. Calculate N after the teacher models M.

Challenge a pair of claims

W2. “All blood-carried hormones affect every cell equally.”

W3. “The pulmonary artery is a vein because its blood is oxygen-poor.”

State what the calculation cannot show

Warm-up N has greater volume output than M.

Does that alone prove a person is healthier?

Before you begin

Which practice will give a useful record of learning?

- A** Record the route and support; answer first; use feedback afterwards.
- B** Convert the total directly into a predicted GCSE grade.
- C** Copy the worked answers while completing the checkpoint.

Set up your selected checkpoint route

Use the shared source page and Q1–Q5.

All routes have the same 20 core marks.

Independent checkpoint: 18 minutes

Answer Q1–Q5. Show units and causal links.

If stuck, move on and return.

Feedback starts now

Pause or collect the first attempt.

Use item evidence to choose one correction.

Audience: identify the next move

L1. Share one correction with an adult or partner.

Say which mechanism or calculation it improves.

Finish with a targeted action

E1. Name one idea you can now explain.

E2. State one practice that would improve your next attempt.

Store the evidence, not a label

Keep the attempt, route, support and correction together.